Lecture 9

Weyl integration and character calculations

Objectives. *A* class function *on a group is a function that is constant on conjugacy classes. On a compact connected group, a class function is determined by its values on a maximal torus. This lecture shows that the Haar integral of a class function can also be computed on the torus. The density is a product over root planes. A factor* $1/|W|$ *corrects for repeated conjugates. We then use the formula to check character norms and tensor-product multiplicities.*

By the end of this lecture you will be able to:

- State the Weyl integration formula with its measure conventions and its imported steps.
- Derive the density $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$ and the factor $1/|W|$.
- Compute class integrals on $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ and compare them.
- Check character norms and multiplicities with the Weyl character formula on T.

9.1 Measures, roots and the statement

Throughout, K is a compact connected Lie group and $T \subset K$ is a maximal torus. Their Lie algebras are $\mathfrak{t} \subset \mathfrak{k}$. By Lecture 6, every element of K is conjugate to an element of T. Also by Lecture 6, the group $W = N_K(T)/T$ is finite and acts on $X^*(T)$ as the Weyl group of the root system. Fix an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$ and set

$$\mathfrak{m} = \mathfrak{t}^{\perp} \subset \mathfrak{k}.$$

For $t \in T$ the map Ad_t is orthogonal and fixes $\mathfrak{t}$ pointwise. Hence Ad_t preserves $\mathfrak{m}$, and $\mathfrak{m} \cong \mathfrak{k}/\mathfrak{t}$ as representations of T.

We use three probability measures.

- dk is the normalised Haar measure on K (Lecture 5).
- dt is the normalised Haar measure on T.
- $d(gT)$ is the unique K-invariant probability measure on K/T.

The existence and uniqueness of $d(gT)$ are cited together with the theorem below. All three measures are given by smooth positive densities.

We keep the real weights and real roots of Lecture 6 and Lecture 7. For $\ell \in X^*(T)$, write t^ℓ for the value at t of the character of T with weight ℓ. Thus $t^\ell = e^{i\ell(H)}$ when $t = \exp H$ with $H \in \mathfrak{t}$, and $t^\ell t^{\ell'} = t^{\ell+\ell'}$.

With this convention, the root decomposition is

$$\mathfrak{t}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \mathrm{Ad}_t E = t^\alpha E \quad (E \in \mathfrak{g}_\alpha,\ t \in T). \tag{9.1}$$

Each $\mathfrak{g}_\alpha$ is a complex line. Each root is a weight of the adjoint representation of K. So $t \mapsto t^\alpha$ is a character of T, that is, $\alpha \in X^*(T)$. If $t = \exp H$ with $H \in \mathfrak{t}$, then $t^\alpha = e^{i\alpha(H)}$ with $\alpha(H) \in \mathbb{R}$.

Fix a positive system R^+ and let ρ be half the sum of the positive roots. In general $\rho \notin X^*(T)$. § 9.4 shows how to avoid ρ as a function on T.

The Lie Algebras course states the Weyl character formula and the denominator identity for complex-linear weights on $\mathfrak{t}_{\mathbb{C}}$. In that course, a real weight ℓ appears as the complex weight $i\ell$, a root α as $i\alpha$, and ρ as $i\rho$. A complex weight $\mu = i\ell$ has the exponential $e^{\mu(H)} = e^{i\ell(H)} = t^\ell$ at $t = \exp H$. The map $\ell \mapsto i\ell$ is additive and commutes with W. So those identities keep their form when we write them with real exponents. They then hold in the ring $\mathbb{Z}[X^*(T)]$ of finite integer combinations of formal exponentials e^ℓ, where $e^\ell e^{\ell'} = e^{\ell+\ell'}$. Evaluating e^ℓ at t gives t^ℓ.

Definition 9.1. An element $t \in T$ is *regular* if $t^\alpha \neq 1$ for every $\alpha \in R$. The regular elements of T form T_{reg}. The regular elements of K form $K_{\mathrm{reg}} = \{gtg^{-1} : g \in K,\ t \in T_{\mathrm{reg}}\}$.

The conjugation map is

$$\psi \colon K/T \times T \to K, \qquad \psi(gT, t) = gtg^{-1}. \tag{9.2}$$

It is well defined because T is abelian. The domain and the codomain of ψ both have dimension $\dim K$.

Theorem 9.2 (Weyl integration formula)**.** *Let K, T, W and the measures be as above. Put*

$$J(t) = \det\bigl(1 - \mathrm{Ad}_t|_{\mathfrak{m}}\bigr) \qquad (t \in T).$$

Then $J \geq 0$, and for every continuous $f \colon K \to \mathbb{C}$,

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T J(t) \int_{K/T} f(gtg^{-1})\,d(gT)\,dt. \tag{9.3}$$

If f is a class function, then

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T f(t)\,J(t)\,dt. \tag{9.4}$$

The theorem is cited from H §11.6, BD IV.1 (1.11) and Knapp, Beyond, VIII.5. We prove the formula below from two imported analytic steps and the existence of $d(gT)$. The class-function version follows from the general formula (9.3). Indeed $f(gtg^{-1}) = f(t)$, and $d(gT)$ has total mass one.

We prove the formula by changing variables along ψ. Every element of K has the form gtg^{-1}. The Jacobian of ψ is a constant multiple of J, and the constant is fixed at the end. A regular

element has $|W|$ preimages, and this gives the factor $1/|W|$. The remaining elements form a set of Haar measure zero.

For example, take $K = \mathrm{SU}(2)$ and $t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. The space K/T is the sphere S of Lecture 3, and the coset gT corresponds to $\xi = g\,\mathrm{diag}(1,-1)g^{-1}$. Then

$$g t_\theta g^{-1} = \cos\theta\, I + i \sin\theta\, \xi.$$

The regular elements of T are the t_θ with $\theta \notin \pi\mathbb{Z}$, and K_{reg} is $\mathrm{SU}(2)$ without $\pm I$. So the remaining elements are two points. A regular element has the two preimages (ξ, θ) and $(-\xi, 2\pi - \theta)$.

9.2 The root-plane Jacobian and the factor $|W|$

Lemma 9.3 (Root planes). *For $\alpha \in R^+$ put $\mathfrak{m}_\alpha = \mathfrak{k} \cap (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha})$. Then $\mathfrak{m} = \bigoplus_{\alpha \in R^+} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has a real basis consisting of X_α and Y_α, in which*

$$\mathrm{Ad}_t\,|_{\mathfrak{m}_\alpha} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \text{where } t^\alpha = e^{i\theta}.$$

Consequently, writing $t^\alpha = e^{i\theta_\alpha}$,

$$J(t) = \prod_{\alpha \in R^+} \left|1 - t^\alpha\right|^2 = \prod_{\alpha \in R} (1 - t^\alpha) = \prod_{\alpha \in R^+} 4 \sin^2 \frac{\theta_\alpha}{2}. \tag{9.5}$$

Proof. Let σ be complex conjugation on $\mathfrak{k}_{\mathbb{C}}$ with respect to $\mathfrak{k}$. It commutes with the real map Ad_t. For $E \in \mathfrak{g}_\alpha$ we get

$$\mathrm{Ad}_t(\sigma E) = \sigma(t^\alpha E) = \overline{t^\alpha}\,\sigma E = t^{-\alpha}\sigma E,$$

because $|t^\alpha| = 1$. So $\sigma \mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$. Choose $E \neq 0$ in $\mathfrak{g}_\alpha$ and set

$$X_\alpha = E + \sigma E, \qquad Y_\alpha = i(E - \sigma E).$$

Both are fixed by σ, so they lie in $\mathfrak{k}$. An element $aE + b\,\sigma E$ is fixed by σ exactly when $b = \bar a$. If $a = x + iy$, then $aE + \bar a\,\sigma E = xX_\alpha + yY_\alpha$. Hence X_α and Y_α form a basis of $\mathfrak{m}_\alpha$. With $t^\alpha = e^{i\theta}$,

$$\begin{aligned} \mathrm{Ad}_t\, X_\alpha &= e^{i\theta}E + e^{-i\theta}\sigma E = \cos\theta\, X_\alpha + \sin\theta\, Y_\alpha, \\ \mathrm{Ad}_t\, Y_\alpha &= ie^{i\theta}E - ie^{-i\theta}\sigma E = -\sin\theta\, X_\alpha + \cos\theta\, Y_\alpha. \end{aligned}$$

Taking σ-fixed points in (9.1) gives $\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha>0} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ is orthogonal to $\mathfrak{t}$. To see this, pick $H \in \mathfrak{t}$ with $\alpha(H) \neq 0$. Then $[H, X_\alpha] = \alpha(H) Y_\alpha$ and $[H, Y_\alpha] = -\alpha(H) X_\alpha$, so $\mathfrak{m}_\alpha \subset [\mathfrak{t}, \mathfrak{k}]$. Invariance of the inner product gives $\langle [H, X], H' \rangle = -\langle X, [H, H'] \rangle = 0$ for $H' \in \mathfrak{t}$. Counting dimensions, $\mathfrak{m} = \bigoplus_{\alpha>0} \mathfrak{m}_\alpha$.

For one plane,

$$\det \begin{pmatrix} 1 - \cos\theta & \sin\theta \\ -\sin\theta & 1 - \cos\theta \end{pmatrix} = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = |1 - e^{i\theta}|^2.$$

This equals $4\sin^2(\theta/2)$. Multiply over $\alpha \in R^+$. Finally $(1 - t^\alpha)(1 - t^{-\alpha}) = |1 - t^\alpha|^2$, since $t^{-\alpha} = \overline{t^\alpha}$. ∎

Thus $J \geq 0$, and $J(t) > 0$ exactly when $t \in T_{\mathrm{reg}}$. Also $J(t^{-1}) = J(t)$.

For $\mathrm{U}(n)$, with $t = \mathrm{diag}(z_1, \dots, z_n)$, the roots of Lecture 6 give the density $\prod_{j<k} |z_j - z_k|^2$. It vanishes when two eigenvalues coincide. So a Haar-random unitary matrix is unlikely to have two close eigenvalues.

Next we write the differential of ψ in translated coordinates. For $X \in \mathfrak{m}$ and $H \in \mathfrak{t}$ put

$$L_{(gT,t)}(X,H) = \frac{d}{ds}\Big|_{s=0} \psi(gT,t)^{-1}\, \psi\big(g\exp(sX)T,\ t\exp(sH)\big) \in \mathfrak{k}.$$

We identify the tangent space of K/T at gT with $\mathfrak{m}$ by $X \mapsto \frac{d}{ds}|_{s=0}\, g\exp(sX)T$. We identify the tangent spaces of T at t and of K at x with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation. In these identifications, each invariant density is given at every point by its density at the base point. Let $\kappa > 0$ be the ratio of the base-point density of dk to the product of the base-point densities of $d(gT)$ and dt, all evaluated on orthonormal bases. Hence the Jacobian of ψ relative to $d(gT)\,dt$ and dk equals $\kappa\,|\det L_{(gT,t)}|$. Here the determinant is taken with respect to orthonormal bases of $\mathfrak{m} \oplus \mathfrak{t}$ and $\mathfrak{k}$.

The value $|\det L_{(gT,t)}|$ does not depend on the representative g of the coset gT. Indeed, replacing g by gs with $s \in T$ composes L with $\mathrm{Ad}_s|_{\mathfrak{m}}$. This map has determinant 1.

The next proposition is the Jacobian computation of Lecture 6, now at a general point. The map ψ is the map Φ there.

Proposition 9.4. *For all $g \in K$ and $t \in T$, $|\det L_{(gT,t)}| = J(t)$.*

Proof. Multiplying out,

$$\psi(gT,t)^{-1}\psi(ge^{sX}T, te^{sH}) = gt^{-1}e^{sX}t\,e^{sH}e^{-sX}g^{-1} = g\ \exp(s\,\mathrm{Ad}_{t^{-1}}X)\exp(sH)\exp(-sX)\,g^{-1}.$$

The velocity at $s = 0$ of a product of one-parameter subgroups is the sum of their velocities (Lecture 2). Hence

$$L_{(gT,t)}(X,H) = \mathrm{Ad}_g\big((\mathrm{Ad}_{t^{-1}} - 1)X + H\big).$$

Relative to $\mathfrak{k} = \mathfrak{m} \oplus \mathfrak{t}$ this is Ad_g composed with the block map

$$\begin{pmatrix} \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}} - 1 & 0 \\ 0 & 1_{\mathfrak{t}} \end{pmatrix}.$$

The map Ad_g is orthogonal, so $|\det \mathrm{Ad}_g| = 1$. Therefore $|\det L| = |\det(1 - \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}})| = J(t^{-1}) = J(t)$. ∎

If K is nonabelian, then $J(t)$ tends to zero as t approaches a central element. Indeed, every root character is 1 at that element. If K is a torus, the root product is empty and J is identically 1.

Proposition 9.5. *(a) For $t \in T$, $\dim Z_K(t) = \dim T + 2\,\#\{\alpha \in R^+ : t^\alpha = 1\}$. Hence $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.*

(b) If $t, t' \in T_{\mathrm{reg}}$ and $gtg^{-1} = t'$, then $g \in N_K(T)$.

(c) Every $x \in K_{\mathrm{reg}}$ has exactly $|W|$ preimages under ψ.

Proof. (a) The Lie algebra of the closed subgroup $Z_K(t)$ is $\{X \in \mathfrak{k} : \mathrm{Ad}_t X = X\}$ (Lecture 3). A plane rotation by θ fixes a nonzero vector only when $e^{i\theta} = 1$. By the root-plane lemma (Lemma 9.3), the fixed space of Ad_t is $\mathfrak{t} \oplus \bigoplus_{t^\alpha = 1} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has dimension 2, which gives the dimension formula. Centraliser dimensions are unchanged by conjugation. So if t is conjugate to a regular t', then t is regular.

Now let $(gT, t) \in \psi^{-1}(K_{\mathrm{reg}})$. Then gtg^{-1} is conjugate to some $t' \in T_{\mathrm{reg}}$, so t is conjugate to t', and hence $t \in T_{\mathrm{reg}}$. Conversely, $\psi(gT, t) \in K_{\mathrm{reg}}$ for every $t \in T_{\mathrm{reg}}$, by the definition of K_{reg}. These two inclusions give $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.

(b) By the dimension formula in (a), the identity component $Z_K(t')_0$ has Lie algebra $\mathfrak{t}$. It contains the connected group T with the same Lie algebra, so $Z_K(t')_0 = T$ (Lecture 2). The torus gTg^{-1} is connected and centralises $gtg^{-1} = t'$. So $gTg^{-1} \subset T$. Then $\mathrm{Ad}_g \mathfrak{t} \subset \mathfrak{t}$, with equality by dimension. Hence $gTg^{-1} \supset \exp \mathfrak{t} = T$, and $g \in N_K(T)$.

(c) Let $x = \psi(gT, t)$ with t regular. Suppose $\psi(hT, s) = x$. By the description of $\psi^{-1}(K_{\mathrm{reg}})$ in (a), s is regular. Put $n = g^{-1}h$. Then $nsn^{-1} = t$. Since s and t are regular, the normaliser statement (b) gives $n \in N_K(T)$. Thus the preimages are the pairs $(gnT,\ n^{-1}tn)$ with $nT \in W$. Each such pair is a preimage. Different classes nT give different cosets gnT. So there are exactly $|W|$ preimages. ∎

Lemma 9.6. $\displaystyle\int_T J(t)\, dt = |W|.$

Proof. The Weyl denominator identity [I] holds in $\mathbb{Z}[X^*(T)]$:

$$\prod_{\alpha > 0} (1 - e^{-\alpha}) = \sum_{w \in W} \mathrm{sgn}(w)\, e^{w\rho - \rho}.$$

Here $\rho - w\rho$ is a sum of positive roots. So $w\rho - \rho \in X^*(T)$ even when $\rho \notin X^*(T)$. Evaluate both sides at t, and write $D(t)$ for the value of the left side. Then $|D(t)|^2 = J(t)$ by (9.5), so

$$J(t) = \sum_{w, w' \in W} \mathrm{sgn}(w)\, \mathrm{sgn}(w')\, t^{w\rho - w'\rho}.$$

Distinct characters of T are orthonormal in $L^2(T)$ (Lecture 5). Thus $\int_T t^\ell\, dt = 0$ for $\ell \neq 0$. Since ρ is regular [I], $w\rho = w'\rho$ only when $w = w'$. Hence only the diagonal terms survive, and the integral is $|W|$. ∎

The proof of the formula uses two imported analytic steps. The section on sources and hypotheses at the end of the lecture derives them from the cited results.

(C1) The set K_{reg} is open, and $K \setminus K_{\mathrm{reg}}$ has Haar measure zero. The null-set claim is cited from BD V.2, Proposition (2.6), together with Lee, Corollary 6.11.

(C2) Let M, N be compact manifolds of equal dimension. Give them finite measures defined by smooth positive densities. Let $\psi\colon M \to N$ be smooth and $U \subset N$ open. Assume $N \setminus U$ and $M \setminus \psi^{-1}(U)$ are null. Assume $\psi\colon \psi^{-1}(U) \to U$ is a surjective local diffeomorphism whose fibres all have d points. Then for continuous F on N,

$$\int_M F(\psi(x))\, J_\psi(x)\, dx = d \int_N F(y)\, dy,$$

where J_ψ is the Jacobian of ψ relative to the two densities. We derive this from Lee, Chapter 16, Propositions 16.40–16.42.

The second step is the change of variables for a map that covers its target d times. For $z \mapsto z^2$ on the circle, we have $d = 2$ and the Jacobian is 2, and

$$\frac{1}{2\pi}\int_0^{2\pi} F(e^{2i\theta})\,2\,d\theta = 2\cdot\frac{1}{2\pi}\int_0^{2\pi} F(e^{i\varphi})\,d\varphi.$$

Proof of the Weyl integration formula (Theorem 9.2) from (C1) and (C2). Null sets. The set $K\setminus K_{\mathrm{reg}}$ is null by the regular-set import (C1). The complement of $\psi^{-1}(K_{\mathrm{reg}})$ is $K/T\times(T\setminus T_{\mathrm{reg}})$. Now $T\setminus T_{\mathrm{reg}}$ is the union of the kernels T_α of $t\mapsto t^\alpha$. Each T_α has infinitely many cosets, because t^α maps the connected group T onto the circle. The cosets are disjoint and have equal Haar measure. Since T has finite measure, T_α is null.

Covering. On $K/T\times T_{\mathrm{reg}}$ the Jacobian is $\kappa J(t)>0$, so ψ is a local diffeomorphism there. By the fibre count in Proposition 9.5, its fibres have $|W|$ points.

Change of variables. Both $K/T\times T$ and K are compact. The change-of-variables import (C2) gives

$$\kappa\int_T J(t)\int_{K/T} f(gtg^{-1})\,d(gT)\,dt = |W|\int_K f(k)\,dk.$$

Take $f=1$. Then $\kappa\int_T J\,dt = |W|$. Since $\int_T J\,dt = |W|$ by Lemma 9.6, we get $\kappa = 1$. This is (9.3). ∎

9.3 The model: class integrals on $\mathrm{SU}(2)$

Let $K=\mathrm{SU}(2)$ with maximal torus $T=\{t_\theta\}$, where

$$t_\theta = \begin{pmatrix} e^{i\theta} & 0\\ 0 & e^{-i\theta}\end{pmatrix},\qquad dt = \frac{d\theta}{2\pi}\ \text{ on } [0,2\pi).$$

Use the invariant inner product $\langle X,Y\rangle = -\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. The plane $\mathfrak{m}$ has basis

$$B=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix},\qquad C=\begin{pmatrix}0 & i\\ i & 0\end{pmatrix}.$$

A direct multiplication gives

$$t_\theta B t_\theta^{-1} = \begin{pmatrix} 0 & e^{2i\theta}\\ -e^{-2i\theta} & 0\end{pmatrix} = \cos 2\theta\, B + \sin 2\theta\, C,\qquad t_\theta C t_\theta^{-1} = -\sin 2\theta\, B + \cos 2\theta\, C.$$

So $t_\theta^\alpha = e^{2i\theta}$ and

$$J(t_\theta) = |1-e^{2i\theta}|^2 = 4\sin^2\theta.$$

The Weyl group has order 2. Its nontrivial element is represented by

$$n=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix},\qquad n t_\theta n^{-1} = t_{-\theta}.$$

Now (9.4) gives, for a continuous class function f,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\,4\sin^2\theta\,d\theta = \frac{1}{\pi}\int_0^{2\pi} f(t_\theta)\sin^2\theta\,d\theta.$$

Since $f(t_{2\pi-\theta}) = f(t_{-\theta}) = f(t_\theta)$ and $\sin^2$ is symmetric about π,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta. \tag{9.6}$$

Every $g \in \mathrm{SU}(2)$ has eigenvalues $e^{\pm i\theta}$ for a unique $\theta \in [0,\pi]$, and $\operatorname{tr} g = 2\cos\theta$. So a class function is a function of $\theta \in [0,\pi]$. For $f = 1$ the right side is $\frac{2}{\pi}\cdot\frac{\pi}{2} = 1$.

Exercise 9.2 shows what this density means on S^3. For $0 < \theta < \pi$, the elements with trace $2\cos\theta$ form a 2-sphere of radius $\sin\theta$, the conjugacy class of t_θ. Its area is proportional to $\sin^2\theta$. At $\theta = 0$ and at $\theta = \pi$ the class is the single point I or $-I$. So $J(t)$ measures the size of the class of t. Elements near $\pm I$ are rare, since those classes are small.

Character norms. Let a representation V decompose as $\bigoplus_i m_i V_i$, with the V_i irreducible and pairwise inequivalent. This is possible by complete reducibility (Lecture 5). Orthonormality of characters (Lecture 5) gives $\int_K \chi_V\overline{\chi_i}\,dk = m_i$ and $\int_K |\chi_V|^2\,dk = \sum_i m_i^2$. So a character of norm one is irreducible.

By the Weyl character formula [I], the representation V_m of dimension $m+1$ has

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin(m+1)\theta}{\sin\theta} \qquad (\theta \notin \pi\mathbb{Z}).$$

Then (9.6) gives

$$\int_{\mathrm{SU}(2)} \chi_m\overline{\chi_n}\,dk = \frac{2}{\pi}\int_0^\pi \sin(m+1)\theta\,\sin(n+1)\theta\,d\theta = \delta_{mn}.$$

The factor $\sin^2\theta$ cancels both denominators. This confirms that each V_m is irreducible and that the representations V_m are pairwise inequivalent.

A tensor-product multiplicity. The multiplicity of V_n in V is $\int_K \chi_V\overline{\chi_n}\,dk$. Take $V = V_1^{\otimes 4}$, so $\chi_V(t_\theta) = (2\cos\theta)^4$. For $n = 0$ we need the Wallis integrals

$$\int_0^\pi \cos^4\theta\,d\theta = \frac{3\pi}{8}, \qquad \int_0^\pi \cos^6\theta\,d\theta = \frac{5\pi}{16}.$$

Write $\sin^2 = 1 - \cos^2$. Then

$$\int_0^\pi \cos^4\theta\sin^2\theta\,d\theta = \frac{3\pi}{8} - \frac{5\pi}{16} = \frac{\pi}{16}, \qquad \dim V^{\mathrm{SU}(2)} = \frac{2}{\pi}\cdot 16\cdot\frac{\pi}{16} = 2.$$

So the trivial representation V_0 occurs twice in $V_1^{\otimes 4}$. Exercise 9.3 finds the full decomposition from weights and checks the multiplicity of V_2 by integration.

Remark 9.7 (Indicator functions)**.** Probabilities of conjugation-invariant sets need discontinuous f. Let $c\colon K \to \mathbb{R}$ be a continuous class function and put $E = \{k \in K : c(k) \geq 0\}$. Then

$$\int_K 1_E\,dk = \frac{1}{|W|}\int_T 1_E(t)\,J(t)\,dt.$$

To prove this, note that the Haar integral on $C(K)$ extends to bounded measurable functions. This is the Haar measure mentioned in Lecture 5, and the same holds on T. We import one

fact about it, the dominated convergence theorem. If measurable functions f_n converge pointwise to f and $|f_n| \le g$ for an integrable g, then $\int f_n \to \int f$.

For $n \ge 1$ let $h_n : \mathbb{R} \to [0,1]$ be continuous with

$$h_n(x) = 1 \ \ (x \ge 0), \qquad h_n(x) = 0 \ \ (x \le -1/n), \qquad h_n \text{ linear on } [-1/n, 0].$$

Each $h_n \circ c$ is a continuous class function. For every k, the values $h_n(c(k))$ decrease to the indicator $1_E(k)$. All these functions are bounded by 1, and J is bounded on T. Applying dominated convergence to both sides of the class-function formula (9.4) gives the claim.

9.4 Characters on the torus and nonsimply connected groups

Let $\lambda \in X^*(T)$ be dominant. By the classification theorem of Lecture 8, K has an irreducible representation V_λ with highest weight λ. Write χ_λ for its character.

Since K is connected, V_λ is determined by its differential. The complex-linear extension of the differential makes V_λ an irreducible $\mathfrak{k}_{\mathbb{C}}$-module, with complex highest weight $i\lambda$. The centre of $\mathfrak{k}_{\mathbb{C}}$ acts by a scalar. Hence the semisimple Weyl character formula [I] applies to the derived algebra, with the central weight retained. Written with real exponents, as explained in § 9.1, the resulting formula is an identity in $\mathbb{Z}[X^*(T)]$. We use it in the form

$$\chi_\lambda(t)\, D(t) = \sum_{w \in W} \operatorname{sgn}(w)\, t^{w(\lambda+\rho)-\rho}, \qquad D(t) = \prod_{\alpha > 0} (1 - t^{-\alpha}). \tag{9.7}$$

Each exponent equals $w\lambda - (\rho - w\rho)$, so it lies in $X^*(T)$. Thus (9.7) is an identity of functions on T for every K. If $\rho \in X^*(T)$, multiplying by t^ρ gives the usual quotient form.

Proposition 9.8. *Let $\lambda, \mu \in X^*(T)$ be dominant. Then $\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \delta_{\lambda\mu}$.*

Proof. Since $|D|^2 = J$, (9.4) gives

$$\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \frac{1}{|W|} \sum_{w,w'} \operatorname{sgn}(ww') \int_T t^{w(\lambda+\rho) - w'(\mu+\rho)}\, dt.$$

A term survives only if $w(\lambda+\rho) = w'(\mu+\rho)$. Both $\lambda+\rho$ and $\mu+\rho$ are strictly dominant. Each W-orbit meets the dominant chamber exactly once, and strictly dominant weights have trivial stabiliser [I]. So a term survives only if $\lambda = \mu$ and $w = w'$. There are $|W|$ such terms. ∎

This agrees with Schur orthogonality. It is a consistency check, not a new proof.

Comparison with SO(3). Let $T = \{R_\varphi\}$, where R_φ is the rotation through φ about the z-axis. Exercise 9.4 shows that Ad_{R_φ} rotates $\mathfrak{m}$ through φ. So $R_\varphi^\alpha = e^{i\varphi}$, $J(R_\varphi) = 4\sin^2(\varphi/2)$ and $|W| = 2$. Formal exponentiation of the weight $\rho = \alpha/2$ would give $e^{i\varphi/2}$. This is not a function on T, because φ and $\varphi + 2\pi$ give the same rotation.

The density J is still defined. Conjugation by the rotation $\mathrm{diag}(1,-1,-1)$ reverses the z-axis, so it maps R_φ to $R_{-\varphi} = R_{2\pi-\varphi}$. Hence a class function satisfies $f(R_{2\pi-\varphi}) = f(R_\varphi)$. The

factor $\sin^2\frac{\varphi}{2}$ is also unchanged under $\varphi \mapsto 2\pi - \varphi$, and $2\sin^2\frac{\varphi}{2} = 1 - \cos\varphi$. Therefore the class-function formula (9.4) gives

$$\int_{\mathrm{SO}(3)} f\,dk = \frac{1}{\pi}\int_0^{2\pi} f(R_\varphi)\sin^2\frac{\varphi}{2}\,d\varphi = \frac{1}{\pi}\int_0^{\pi} f(R_\varphi)(1-\cos\varphi)\,d\varphi. \tag{9.8}$$

We compare this with $\mathrm{SU}(2)$. The image of Haar measure under the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is the Haar measure of $\mathrm{SO}(3)$, by uniqueness (Lecture 5). The element t_θ maps to a rotation through $\pm 2\theta$. So the substitution $\varphi = 2\theta$ turns (9.6) into the first form of (9.8). Exercise 9.4 shows that small rotations are rare.

The irreducible representations of $\mathrm{SO}(3)$ are the V_{2j} (Lecture 8). Here $\lambda = j\alpha$, and (9.7) reads

$$\chi_{2j}(R_\varphi)\,(1 - e^{-i\varphi}) = e^{ij\varphi} - e^{-i(j+1)\varphi}.$$

Hence

$$\int |\chi_{2j}|^2 = \frac{1}{2}\int_T |e^{ij\varphi} - e^{-i(j+1)\varphi}|^2 dt = \frac{1}{2}\cdot 2 = 1.$$

Self-checks

1. Find J for $\mathrm{U}(2)$ with $t = \mathrm{diag}(e^{ia}, e^{ib})$. Is $\rho \in X^*(T)$?
   Answer. There is one positive root, $t^\alpha = e^{i(a-b)}$. So $J = |e^{ia} - e^{ib}|^2 = 4\sin^2\frac{a-b}{2}$. Identify $X^*(T)$ with $\mathbb{Z}^2$, where (p,q) labels the weight ℓ with $t^\ell = e^{i(pa+qb)}$. Then $\alpha = (1,-1)$ and $\rho = (\frac{1}{2}, -\frac{1}{2}) \notin X^*(T)$.
2. What is the Haar probability that $g \in \mathrm{SU}(2)$ has $\operatorname{tr} g \geq 0$?
   Answer. The remark on indicator functions (Remark 9.7) with $c(g) = \operatorname{tr} g$ lets us use (9.6) for this indicator. The condition on the trace means $\theta \in [0, \pi/2]$. The probability is
   $$\frac{2}{\pi}\int_0^{\pi/2} \sin^2\theta\,d\theta = \frac{2}{\pi}\cdot\frac{\pi}{4} = \frac{1}{2}.$$
   This matches the symmetry $g \mapsto -g$.
3. Why can $\int_{\mathrm{SU}(2)} |g_{11}|^2 dk$ not be computed from (9.6) with $f(t_\theta) = |e^{i\theta}|^2$?
   Answer. The function $g \mapsto |g_{11}|^2$ is not a class function. Schur orthogonality (Lecture 5) gives the correct value $1/\dim V_1 = \frac{1}{2}$. Using $f(t_\theta) = 1$ in (9.6) would give the wrong value 1. To compute this integral with the Weyl integration formula, one must use the general form (9.3).

Summary

Let K be a compact connected Lie group with maximal torus T. Every regular element of K has exactly $|W|$ preimages under the conjugation map $K/T \times T \to K$. Relative to the normalised measures, the Jacobian of this map at (gT, t) is $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$. This determinant is a product of root-plane factors $|1 - t^\alpha|^2$ over the positive roots α. The Weyl denominator identity fixes the constant. For $\mathrm{SU}(2)$ and a continuous class function f, the Haar integral of f is $\frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta$. With the character formula in the form (9.7), norms and multiplicities become Fourier calculations on T. This form never needs ρ to be a character.

Sources and hypotheses

The regular set, (C1). BD V.2, Proposition (2.6), covers $K \setminus K_{\mathrm{reg}}$ by the image of a compact manifold of dimension at most $\dim K - 3$. Such an image is null, because a smooth map from a manifold of lower dimension has a null image (Lee, Corollary 6.11). Haar measure has a smooth positive density, so a set that is null in charts has Haar measure zero. BD's regular elements agree with ours by the centraliser-dimension formula of Proposition 9.5.

Openness also follows from that proposition. The map ψ is onto (Lecture 6), and $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$. Hence $K \setminus K_{\mathrm{reg}} = \psi\big(K/T \times (T \setminus T_{\mathrm{reg}})\big)$. The set $T \setminus T_{\mathrm{reg}}$ is a finite union of kernels of characters, so it is closed in T. So $K \setminus K_{\mathrm{reg}}$ is a continuous image of a compact set. Hence it is compact, and therefore closed in K.

The change of variables, (C2). The source is Lee, Chapter 16, Propositions 16.40–16.42. Proposition 16.40 is the absolute-Jacobian rule for pulling back a density in coordinates. Lee defines integrals of densities through charts and a partition of unity. Lee does not state (C2) itself, so we derive it from these results.

First, each point $y \in U$ has an open neighbourhood $O \subset U$ whose inverse image consists of d disjoint open sets, each mapped diffeomorphically onto O. To see this, let $x_1, \dots, x_d$ be the preimages of y. Choose disjoint open sets $V_i \ni x_i$ in $\psi^{-1}(U)$ that ψ maps diffeomorphically onto open sets $\psi(V_i)$. The set $C = M \setminus \bigcup_i V_i$ is closed in the compact manifold M, so it is compact. It contains no preimage of y. So $\psi(C)$ is compact, hence closed, and $y \notin \psi(C)$. Put $O = \bigcap_i \psi(V_i) \setminus \psi(C)$. This set is open and contains y. Every preimage of a point of O lies outside C, hence in some V_i. So the inverse image of O is the disjoint union of the sets $V_i \cap \psi^{-1}(O)$. Since $O \subset \psi(V_i)$, the map ψ sends each of these sets diffeomorphically onto O.

Thus each point of U has a neighbourhood that is evenly covered by d sheets, as in the definition of a covering map in Lecture 4. The domain need not be connected. For SU(2) it has two components, and each maps diffeomorphically onto K_{reg}.

Next, suppose F is supported in such a set O. Apply the change-of-variables rule on each of the d sets $V_i \cap \psi^{-1}(O)$ and add the results. A partition of unity on U subordinate to these sets O gives the general case: the partition is countable; sum the pieces by dominated convergence, since the partial sums are bounded by $|F \circ \psi| J_\psi$ and $|F|$. The null sets $N \setminus U$ and $M \setminus \psi^{-1}(U)$ do not contribute. BD I.5, (5.19), states the oriented version through the mapping degree.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., §11.6.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1, V.2 and VI.1.
- Knapp Beyond: *Lie Groups Beyond an Introduction*, VIII.5.
- Lee: *Introduction to Smooth Manifolds*, 2nd ed., Corollary 6.11, and Chapter 16, Propositions 16.40–16.42.